FI FISHINTEL GLOBAL

ENVIRONMENTAL RECONSTRUCTION REPORT

Decision-Time Environmental Reconstruction

Capsize of Tug During Ship Assist Towage Operation

INCIDENT DATE	24 February 2023
REPORT REFERENCE	FLAGSHIP005
DECISION FRAME	15:00 UTC
REPORT VERSION	v1.0 — Chain v1.0
VESSEL CATEGORY	Twin screw workboat / tug
CLASSIFICATION	Public — Anonymised
OPERATIONAL CONTEXT	Towing — 2 POB
HAZARD	Capsize / stability failure (girting)

PIPELINE VERIFIED — EVIDENCE CHAIN VERIFIED — 26 ARTIFACTS SEALED

PUBLIC VERSION — ANONYMISED CASE STUDY — VESSEL AND INDIVIDUALS NOT IDENTIFIED

01

Executive Summary

Incident type: Capsize during ship assist towage. A conventional tug, operating as the stern tug for a passenger vessel navigating a Scottish port channel, girted, capsized, and sank. Both crew members were lost. Classified as a Very Serious Marine Casualty.

Core question: Did environmental conditions contribute to the capsize, and was the operational decision window constrained by weather at any point during the towage?

Method: ERA5 atmospheric reanalysis was extracted across a five-point window (12:00–16:00 UTC) at the incident coordinates and processed through the FishIntel deterministic reconstruction pipeline. This is an atmospheric-only reconstruction; ERA5 wave data is unavailable at this enclosed estuarine location.

Key result: Environmental conditions were operationally benign throughout. At the decision frame (7.3 kt sustained, improving to 6.1 kt by 16:00), the reconstruction classifies risk as LOW and the T_{irr} framework returns `REGIME 1 — REGIME_1_SAFE`. The full environmental decision budget was intact at the moment of capsize. The loss was not environmentally forced.

DECISION SNAPSHOT — 15:00 UTC

LOCATION	DATE	PERSONS ON BOARD
Navigational channel, Greenock, Scotland	24 February 2023	2
FATALITIES	**ERA5 WIND (DECISION FRAME) TIER 1**	**ERA5 WAVE DATA**
2	7.3 kt	N/A — land-masked
PREDICTION TIER 2	**RISK REGIME TIER 2**	**ENVIRONMENTAL CONSTRAINT**
LOW / GO (17.6%)	REGIME 1 — REGIME_1_SAFE	None — full decision budget

02

Incident Overview

Vessel Profile

TYPE	Twin screw workboat / tug
OPERATION	Ship assist towage — stern tug
PERSONS ON BOARD	2 (master and deckhand)
ENGINE STATUS	OPERATIONAL — both engines running at time of capsize
TOWING ARRANGEMENT	Gob rope configuration with towing bridle
HAZARD CLASSIFICATION	CAPSIZE_STABILITY (FIG Step 11, exact match)

On 24 February 2023, the passenger vessel was returning to port after earlier trials, with a pilot embarked. The vessel entered the navigational channel at approximately 15:15. The tug made fast aft as the stern tug. Conditions were unremarkable: the MAIB report records NW Beaufort force 2, good visibility, sea state 2, and high water, slack tidal stream.

The tug was unable to reverse direction to operate directly astern before its weight came onto the towing bridle. The gob rope did not prevent the tug from being towed sideways. The passenger vessel's speed produced towline loads estimated at between two and five times the port's recommended speed range for conventional tug operations. The tug girted, capsized, and sank. Both crew members were lost.

The MAIB investigation identified fourteen safety issues directly contributing to the casualty, and a further six indirectly contributing. The principal factors relevant to this environmental reconstruction are summarised in Section 05 as condensed neutral data statements.

03

Decision-Time Environment

The following chronology presents the operational sequence from the start of the ERA5 extraction window through the capsize, with environmental context from the reconstruction at each stage.

12:00 UTC

Extraction window opens

ERA5 reconstruction begins. Light northerly winds, good visibility, stable pressure.

```
ERA5 12:00: 7.6 kt sustained, 19.6 kt gusts, 1016.0 hPa  Tier 1
```

12:00–15:00 UTC

Environmental conditions improving

Wind and gusts declining steadily across the extraction window. Pressure stable. Visibility good throughout.

```
ERA5 14:00: 7.4 kt sustained, 17.2 kt gusts | Trend: improving  Tier 1
```

15:00 UTC — DECISION FRAME

Primary decision frame

ERA5 decision-frame timestamp. 7.3 kt sustained, 17.1 kt gusts, visibility good (dewpoint spread 7.8°C), pressure 1015.9 hPa. Wind decreasing from 12:00 baseline.

```
ERA5 15:00: 7.3 kt | Prediction: LOW / GO | Decision window: OPEN  Tier 1
```

~15:15 UTC

Passenger vessel enters channel — tug makes fast aft

Passenger vessel enters the navigational channel with pilot embarked. Tug makes fast aft as stern tug. Tow underway.

```
Environmental conditions unchanged from 15:00 decision frame  Tier 3
```

~15:27 UTC

Girting — capsize

Tug unable to reverse direction to operate directly astern before its weight came onto the towing bridle. The gob rope did not prevent the tug from being towed sideways. Tug girted, capsized, and sank. Both crew members were lost.

> No environmental deterioration detected — capsize driven by operational towing forces `Tier 3`

04

Environmental Reconstruction Tier 1

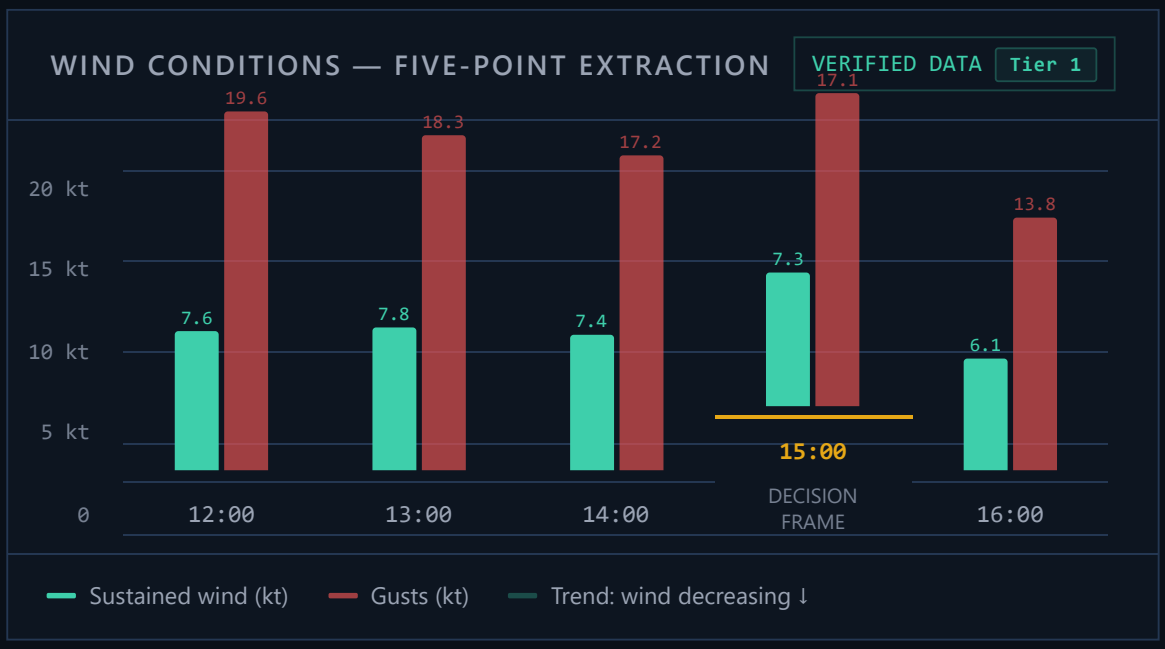
Atmospheric Reconstruction Only. ERA5 wave data is unavailable at this location. The upper Firth of Clyde is land-masked in the ERA5 WAM wave model at 0.5° resolution. The MAIB report states sea state 2 (estimated 0.2 m) — this is a Tier 3 contextual observation, not ERA5-derived evidence.

UTC	WIND (KT)	GUST (KT)	DIRECTION	PRESSURE	TEMP	DEW SPREAD	CLOUD
12:00	7.6	19.6	350°	1016.0	9.6	5.6	50%
13:00	7.8	18.3	355°	1016.0	10.0	5.9	63%
14:00	7.4	17.2	355°	1015.9	10.3	6.8	86%
15:00 ◂ DECISION FRAME	**7.3**	**17.1**	**355°**	**1015.9**	**10.4**	**7.8**	**91%**
16:00	6.1	13.8	357°	1016.1	9.8	7.3	90%

7.3 kt	17.1 kt	1015.9 hPa	10.4 °C
WIND AT DECISION FRAME	GUST AT DECISION FRAME	PRESSURE (MSL)	AIR TEMPERATURE

ENVIRONMENTAL TREND

Wind speed decreased monotonically: 7.6 kt at 12:00 to 6.1 kt at 16:00. Gusts decreased from 19.6 kt to 13.8 kt. Pressure stable at 1016.0–1016.1 hPa. Dewpoint spread increased from 5.6°C to 7.3°C — confirming sustained good visibility. Conditions were becoming more benign, not less.



05

Operational Context Tier 3

The MAIB report identifies fourteen safety issues directly contributing to the casualty and six indirectly contributing (MAIB Section 3). The following are condensed summaries of the principal operational factors. They are presented as neutral data statements. For the full findings, refer to the MAIB report.

MAIB Contributing Factors — Condensed Summaries Tier 3

1. Tug unable to reverse direction before towline loaded
2. Gob rope did not prevent tug being towed sideways
3. Assisted vessel speed exceeded port guideline for conventional tug operations
4. Incomplete information exchange between pilot and tug
5. Watertight integrity compromised by open accommodation hatch

Condensed from MAIB Report 17/2024, Sections 3.1–3.2 (20 safety issues total). Not verbatim. No fault attribution.

NO VESSEL STABILITY DATA

No stability assessment, Wolfson guidance, or freeboard data is available for the tug. The capsize mechanism (girting) is a towing-geometry failure, not a conventional stability-limit exceedance. Environmental conditions are assessed as having no contribution to the capsizing forces.

06

Decision Window Assessment Tier 2

FishIntel's Time-to-Irreversibility (T_{irr}) framework identifies when an operational situation becomes irreversible — the point beyond which no reasonable intervention can change the outcome.

Decision Window — Reversibility Timeline

Time budget from operational commitment to environmental irreversibility at ERA5 baseline conditions.
Decision frame: 15:00 UTC — 27 minutes before capsize.

6 min	33 min	27 min
$T_{PRACTICAL}$	T_{IRR} (EFFECTIVE)	MARGIN (BEST)

Decision Window Findings

RISK REGIME	REGIME 1 – REGIME_1_SAFE
MECHANISM	GRADUAL_DEGRADATION
T_{IRR} (EFFECTIVE)	33 min (2,000,000 ms) — ceiling
ENVIRONMENTAL CONSTRAINT	None — full decision budget available

DECISION FRAME	15:00 UTC — 27 minutes before capsize
ENVIRONMENTAL TREND	Wind decreasing. Conditions improving. No deterioration.

INTERPRETATION

At the ERA5 decision frame, the environmental situation is classified as Regime 1 — Safe. The effective T_{irr} saturates at the regime ceiling (2,000,000 ms), meaning environmental conditions at no point during the operational window approached any irreversibility threshold. The entire decision budget — the time available for recognition, assessment, and intervention — remained fully available to the operational actors. The environment imposed no constraint on the decisions that were made.

07

Risk Analysis Tier 2

TIMESTAMP	RISK SCORE	LEVEL	DECISION
2023-02-24T12:00:00	41.9	MODERATE	CAUTION
2023-02-24T13:00:00	35.1	LOW	ACCEPTABLE
2023-02-24T14:00:00	33.1	LOW	ACCEPTABLE
2023-02-24T15:00:00	33.2	LOW	ACCEPTABLE
2023-02-24T16:00:00	26.9	LOW	ACCEPTABLE

MODERATE CLASSIFICATION — 12:00 UTC TIER 2

The single MODERATE classification at 12:00 UTC is a model-layer sensitivity output driven by the gust-factor penalty (gust ratio 2.59 at absolute gust speed 19.6 kt). This represents gust variability at low absolute wind speeds, not hazardous conditions. By the decision frame (15:00 UTC), all classifications are LOW.

LOW	17.6%	↓	GO
DECISION-FRAME CLASS.	RISK SCORE AT 15:00	RISK TREND (IMPROVING)	ENVIRONMENTAL DECISION

ERA5 ENVIRONMENTAL ASSESSMENT

OPERATIONAL OUTCOME

CAPSIZE

7.3 kt

Sustained wind at decision frame

LOW / SAFE

All parameters within limits — decision budget intact

Girting during ship assist towage

2 fatalities

Both crew members lost — not environmentally forced

08

Key Analytical Finding

PRIMARY FINDING

The environmental conditions were operationally benign. The decision budget was fully intact. The capsize was not environmentally forced.

At the decision frame (15:00 UTC), ERA5 records 7.3 kt sustained wind, 17.1 kt gusts, good visibility, and stable barometric pressure. Wind speed was declining throughout the extraction window — 7.6 kt at 12:00 falling to 6.1 kt at 16:00. The T_{irr} framework classifies the situation as Regime 1 — REGIME_1_SAFE, with the effective T_{irr} at the regime ceiling. No environmental escalation of any kind was detected.

The capsize resulted from compound operational factors: the tug was unable to reverse direction to operate directly astern before its weight came onto the towing bridle, the gob rope did not prevent it from being towed sideways, the passenger vessel's speed produced towline loads estimated at two to five times the port's recommended range for conventional tug operations, and there was incomplete information exchange between the pilot and the tug. No environmental forcing capable of contributing to the capsize was detected.

This reconstruction quantifies what the environment did not do. It did not deteriorate. It did not exceed any operational threshold. It did not reduce the time available for intervention. The entire margin between safe operation and catastrophe was consumed by the operational configuration — not by weather.

This report does not attribute fault or negligence to any individual. Its purpose is to establish, with quantified precision, the environmental context in which the operational decisions were made.

09

Evidence Integrity

This report is a rendering of outputs produced by the FishIntel deterministic reconstruction pipeline. All environmental values are traceable through an auditable cryptographic chain. No manual adjustments or subjective corrections are applied.

EVIDENCE HASH CHAIN — V1.0 (ATMOSPHERIC ONLY)

SOURCE INTEGRITY TIER 1

ERA5 ATMO SEAL	5078b72de9fb6fd08e392265c4fc79981d61dfcd34c3695b1a0f44ae51923daa
ERA5 WAVE SEAL	N/A — land-masked location
UNIT GATE CONFIG	989aadf7b9b1a4acef8606e162015337e75a8ce5fcb4edfb1e0d79f4d59513aa

MODEL INTEGRITY TIER 2

CORE V7 INPUT	e02ee072f61d8a5e2c11c0fa33896f8c1ea6f057e0394564bbf82fcaa69ca8d3
CORE V7 AUDIT	a04a29cece138abb51a27ac800173e7b3e46852ba299b661175cf5a44932d2b0
T_{IRR} HASH	b2b0328d5b06ada181eaa7f646789ddf7e96bdc8835a7b3a5b85830914381af6

CHAIN INTEGRITY

COMBINED HASH	1016154434a0af1e303ba83edafce204af9071bbb3d9b45982be5f8629b3813b
PROTOCOL HASH	a3e76b4ad166c65ec870564e9774acf99469088d82be8e61e5d6f8c878eab230

Methodology

FishIntel Global reconstructs environmental conditions using ECMWF ERA5 atmospheric reanalysis — the same dataset relied upon by meteorological agencies, climate researchers, and aviation safety authorities worldwide. ERA5 provides hourly estimates on a global grid at approximately 31 km horizontal resolution.

All values pass through a unit-gated validation layer enforcing explicit conversion factors and plausibility bounds using Decimal arithmetic. The pipeline is deterministic: identical inputs always produce identical outputs.

This reconstruction is atmospheric-only. ERA5 wave data from the WAM ocean wave model is unavailable at the incident location because the upper Firth of Clyde is classified as land in the ERA5 wave grid at 0.5° resolution. The MAIB observation of sea state 2 (estimated 0.2 m) is cited as contextual information but does not enter the Tier 1 evidence chain.

FishIntel Analytical Architecture

FishIntel operates as a multi-system analytical architecture designed to reconstruct the environmental conditions surrounding maritime incidents and quantify the operational margin before irreversible outcomes occur.

The architecture consists of several interlocking analytical systems:

ARCHITECTURE COMPONENTS

Source Extraction System — converts investigation material and case information into structured machine-readable inputs.

Environmental Reconstruction System — retrieves and reconstructs decision-time environmental conditions using ERA5 atmospheric reanalysis.

Unit Integrity System — validates and canonicalises all physical measurements using explicit conversion factors and Decimal arithmetic.

Risk Inference System — evaluates reconstructed environmental conditions to produce structured risk assessments.

Time-to-Irreversibility (T_{irr}) System — calculates the operational window before intervention becomes ineffective.

Evidence Chain System — generates deterministic seals, hashes, and verification artifacts for every analytical step.

Report Generation System — renders validated outputs into investigation-grade analytical reports.

These systems operate together within a deterministic reconstruction pipeline.

Audited Implementation Modules

The modules listed below are audited implementation components supporting the FishIntel analytical architecture.

MODULE	VERSION	FUNCTION
FIG_CANONICAL.py	1.1.1	Deterministic serialisation and evidence hashing
FIG_HAZARD_CATEGORY.py	1.0.0	Hazard classification mapping

FIG_CASE_CONTEXT.py	1.0.0	Provenance-tracked case context schema
FIG_T_IRR_V1.py	1.0.0	Time-to-Irreversibility analysis engine
FIG_UNIT_GATE.py	1.0.0	Unit conversion and plausibility gating
FIG_SOURCE_SEAL.py	1.0.0	Tamper-evident raw data ingest sealing
FIG_COMBINED_EVIDENCE.py	2.0.0	Combined evidence packaging and hash chain
FIG_CORE_V7_WEAPON.py	7.4	Deterministic risk prediction engine
ERA5_PROCESSOR_V5.py	5.3.0	ERA5 NetCDF extraction and processing

11

Limitations and Caveats

ERA5 RESOLUTION

31 km horizontal, 1-hour temporal. Captures synoptic conditions only. Cannot resolve harbour-scale wind variations.

ATMOSPHERIC ONLY

ERA5 wave data unavailable. Location is land-masked in WAM model. MAIB sea state 2 observation is `Tier 3` contextual only.

NO STABILITY DATA

No stability assessment or Wolfson data available for the tug. Capsize mechanism is girting, not conventional stability failure.

DERIVED COORDINATES

55.951°N, 4.743°W derived from MAIB chartlet and landmarks. ±300 m. Immaterial at ERA5 31 km grid scale.

NO CROSS-VALIDATION

ERA5 only. v1.0 chain (9 components). No independent GFS or observational verification.

WAVE HEIGHT SHIM

Runner and prediction adapter use 0.2 m (estimated from MAIB sea state 2) as a `Tier 3` compatibility input. Does not enter Tier 1 evidence chain.

12

Closing Statement

Not every maritime casualty is driven by weather. In sheltered waters, on a calm day, the environmental reconstruction may confirm what experienced mariners already suspect — that the conditions were benign, and that the loss must be explained by something other than the environment.

In this case, ERA5 atmospheric reanalysis confirms exactly that. Wind speeds were light and declining. Pressure was stable. Visibility was good. The T_{irr} framework classifies the environmental situation as Regime 1 — Safe, with the full decision budget intact at the moment of capsize.

The casualty was driven by the interaction of towing geometry, vessel speed, information exchange, and the physical limitations of the gob rope arrangement. The environment was not the driver of the casualty. The question this reconstruction leaves for investigators, insurers, and regulators is not whether the weather was dangerous — it was not — but how to assess operational risk when the hazard is embedded entirely in the operational configuration itself.

Appendix

Evidence Manifest

26 artifacts. Every numeric value programmatically extracted from these files.

FILENAME	SIZE	DIRECTORY
audit_pack.json	3,686	evidence_flagship005/
case_context.json	2,399	evidence_flagship005/
combined_evidence.json	2,255	evidence_flagship005/
combined_legal_evidence.txt	2,936	evidence_flagship005/
era5_atmo_seal.json	937	evidence_flagship005/
gated_values_1200.json	744	evidence_flagship005/
gated_values_1300.json	744	evidence_flagship005/
gated_values_1400.json	744	evidence_flagship005/
gated_values_1500.json	744	evidence_flagship005/
gated_values_1600.json	743	evidence_flagship005/
hazard_classification.json	241	evidence_flagship005/
prediction_1200.json	3,719	evidence_flagship005/
prediction_1300.json	3,723	evidence_flagship005/
prediction_1400.json	3,728	evidence_flagship005/

prediction_1500.json	3,724	evidence_flagship005/
prediction_1600.json	3,728	evidence_flagship005/
raw_era5_1200.json	1,823	evidence_flagship005/
raw_era5_1300.json	1,824	evidence_flagship005/
raw_era5_1400.json	1,825	evidence_flagship005/
raw_era5_1500.json	1,823	evidence_flagship005/
raw_era5_1600.json	1,826	evidence_flagship005/
tirr_analysis.json	802	evidence_flagship005/
tirr_legal_evidence.txt	2,423	evidence_flagship005/
reconstruction_results.csv	666	reconstruction_results_flagship005/
reconstruction_results.json	3,903	reconstruction_results_flagship005/
reconstruction_summary.txt	338	reconstruction_results_flagship005/

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